

2021

Cybersecurity
INSIDERS

REMOTE WORKING RISKS: Cybersecurity Survey



DARKTRACE

OVERVIEW

As organizations continue to adapt to new working patterns, the need to protect their digital assets from novel and sophisticated threats is more critical than ever. With many employees continuing to work from home and cloud-based collaboration platforms set to become a permanent part of the digital infrastructure, an adaptive and holistic approach to cybersecurity is needed. The Remote Working Risks: Cybersecurity Survey reveals the status of organizations' efforts to secure the dynamic workforce, key challenges and unique security threats faced by organizations, technology gaps and preferences, investment priorities, and more.

As cyber-attacks targeting the remote workforce grow in scale and speed, traditional security tools are failing to keep up. Recognizing this, organizations are increasingly turning to artificial intelligence for adaptability and resilience when protecting their workforce and digital estate.

Key findings include:

- Phishing (77%) and malware (61%) have emerged as the most significant new threat vectors concerning remote workers.
- Virtually all organizations (98%) are concerned about sophisticated impersonation and/or phishing email attacks.
- A majority of organizations 71% confirms the importance of the security tool capability to automatically adapt to the changing digital environment.
- 87% of security professionals agree that signature and rule-based tools are incapable of detecting novel (zero-day) attacks.

We would like to thank [Darktrace](#) for supporting this important research. We hope you find this report informative and helpful as you continue your efforts in protecting your organization against evolving threats and during challenging times.

Thank you,

Holger Schulze



Holger Schulze

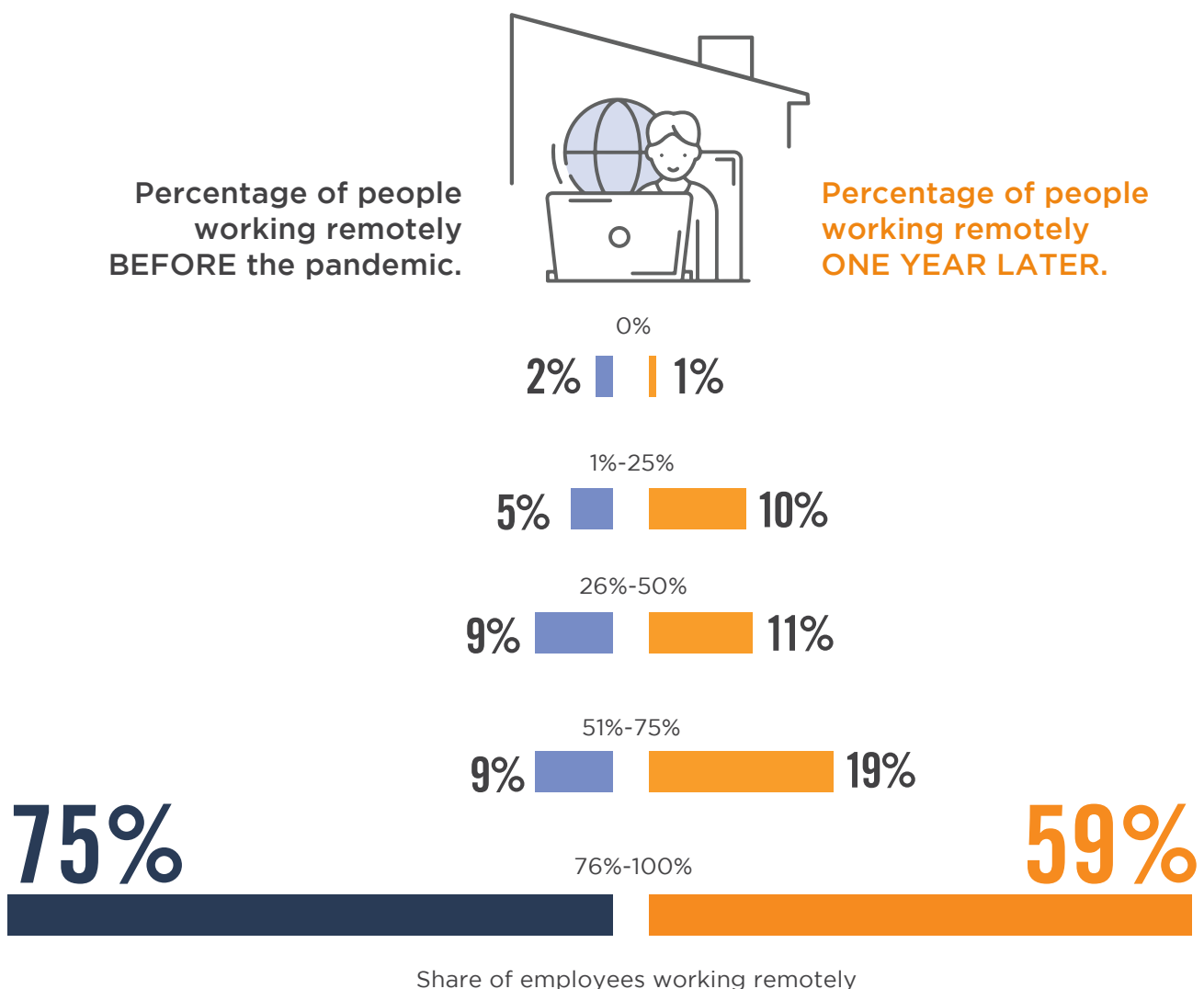
CEO and Founder
Cybersecurity Insiders

Cybersecurity
INSIDERS

DRAMATIC INCREASE IN REMOTE WORKFORCE

In the first year of the COVID-19 pandemic, a majority of organizations made the shift to a remote workforce for almost all employees with 75% of companies having more than three-quarters of their employees work from home. One year into the pandemic, 59% of organizations still have over 75% of their workforce remote. As the threat of the pandemic diminishes in some parts of the world, more employees are returning back to their traditional office environments. We expect a good portion of the current remote workforce to permanently stay remote.

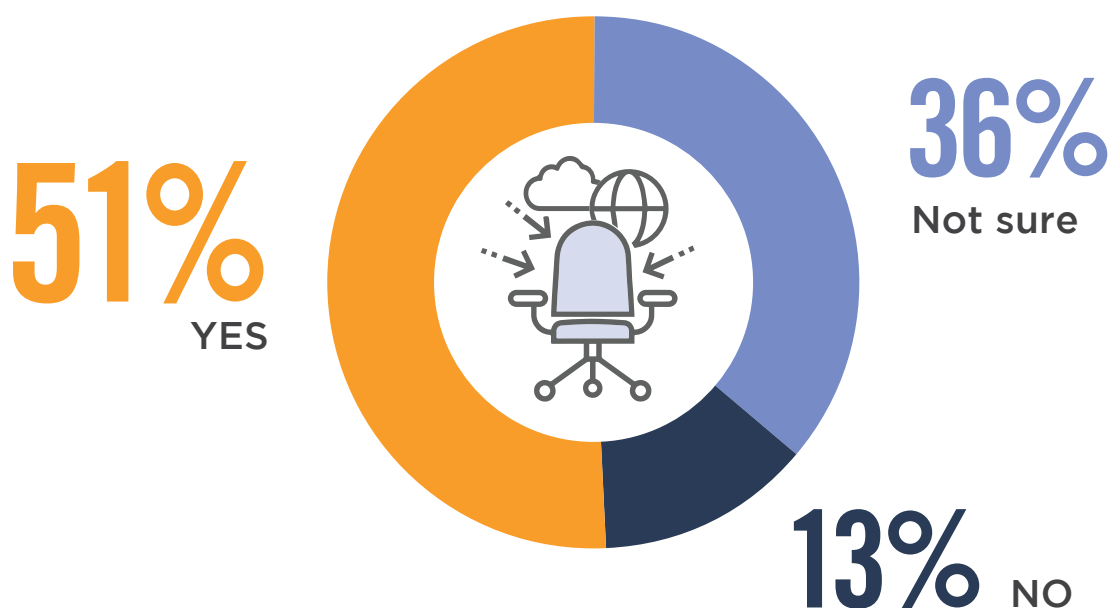
▶ **What percentage of your workforce is working remotely now compared to just before the onset of the COVID-19 pandemic?**



FUTURE REMOTE WORK

More than half of organizations are considering making some positions permanently remote (51%) after the COVID threat passes. Employers are also seeing benefits of remote work, including lower cost and increased productivity for some roles.

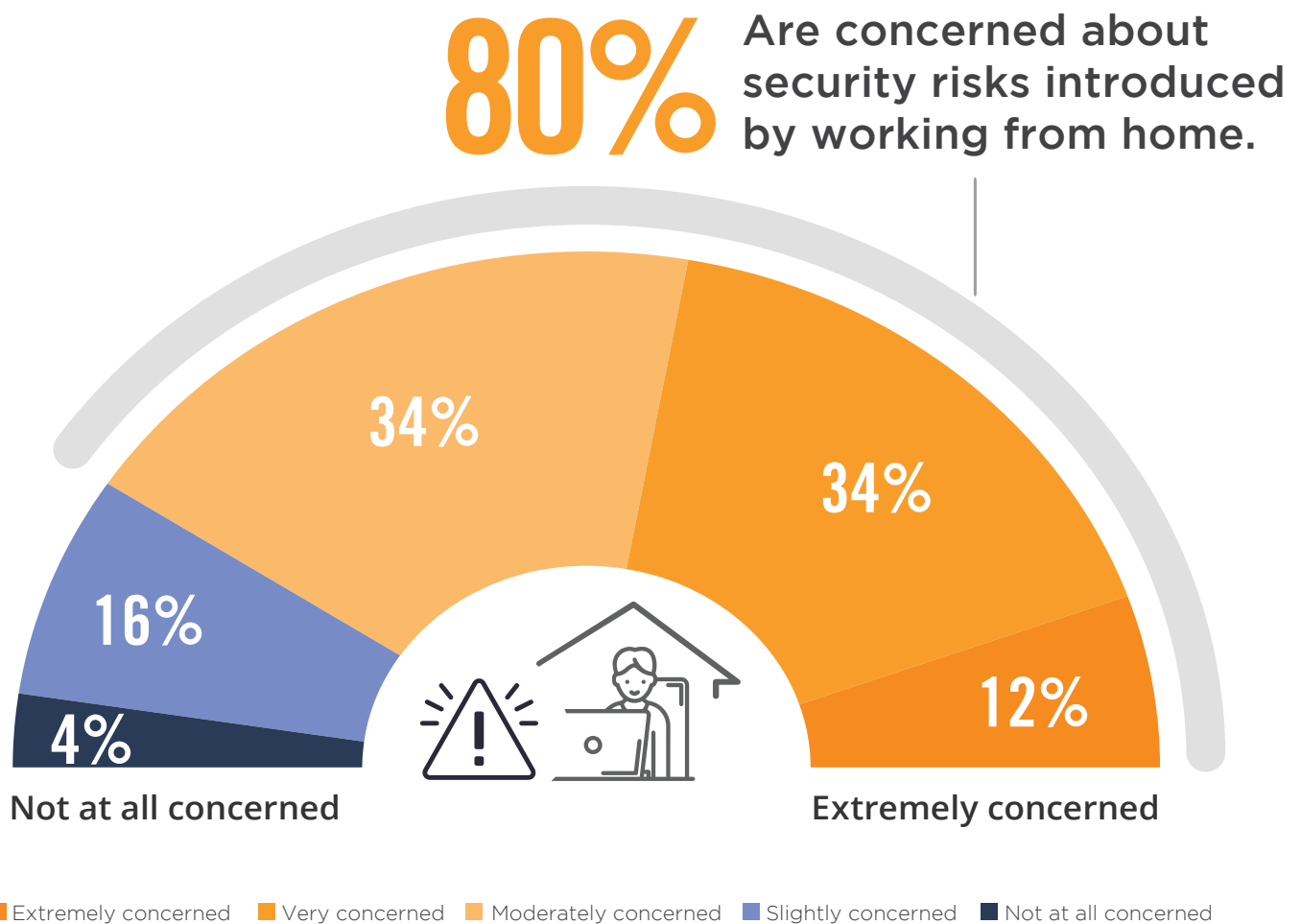
- ▶ **Is your organization considering to make some positions permanently remote (that used to be on-site) after the COVID crisis ends?**



SECURITY RISK CONCERNS

Although organizations have made the shift to remote work, security concerns remain high. Eighty percent of organizations are moderately to extremely concerned about the security risks introduced by people working from home.

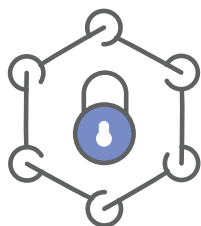
► How concerned are you about the security risks introduced by users working from home?



WIDENED ATTACK SURFACE

A unified, enterprise-wide security solution is critical for protecting the entire digital estate. Security breaches at the endpoint are a source of concern for many organizations as they look to secure their corporate assets. Network access (69%) and BYOD (60%) top the list of security concerns. Slightly less of a concern are managed devices (47%) and SaaS applications, such as Zoom, Slack, etc. (41%).

► What is your organization primarily concerned with securing while employees work remotely?



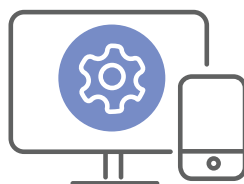
69%

Corporate
network
access



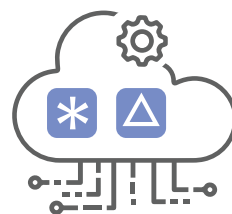
60%

BYOD/
personal
devices



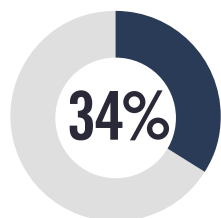
47%

Managed
devices

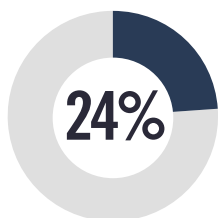


41%

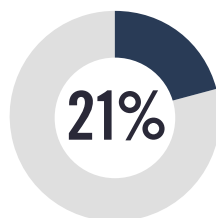
SaaS apps
(Zoom, Slack,
Salesforce, GSuite,
HR platforms, etc.)



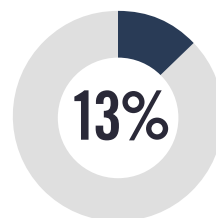
On-premises
apps



The Web



Custom apps



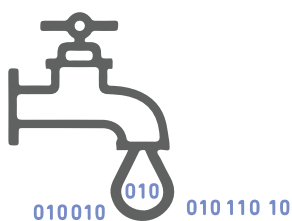
Cloud
environment

IaaS instances 11% | Employee email environment 8%

NEW SECURITY RISKS

With more people now working from home, organizations need to address new security risks created by the remote workforce. Data leaking through endpoints (62%) and users connecting with unmanaged devices are tied at 62% as the biggest new security threats. Access from outside the perimeter (52%) is another key concern followed by maintaining compliance with regulatory requirements (48%).

► What are your concerns about the security risks introduced by new classes of remote users while working from home?



62%

Data leakage to endpoints



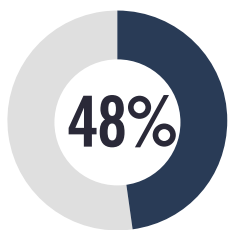
62%

Users connecting with unmanaged devices

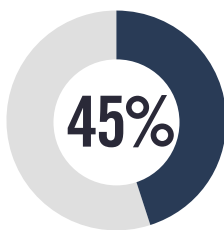


52%

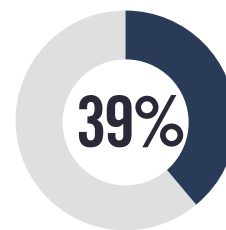
Access from outside the perimeter, meaning less anti-malware protection



Maintaining compliance with regulatory requirements



Loss of visibility of user activity



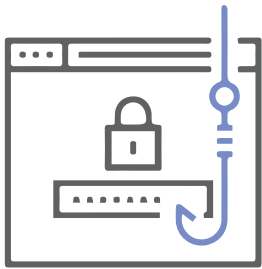
Remote access to core business apps (i.e., email and collaboration)

Password reuse across personal devices 30% | Direct access to cloud-based applications around our IDP 21% | Insider threats 13% | Employees increasingly vulnerable to phishing attacks 8% | Other 8%

THREAT VECTORS

Phishing (71%) and malware (61%) have emerged as the most significant new threat vectors concerning remote work environments.

► What specific threat vectors are you most concerned about with employees working from home?



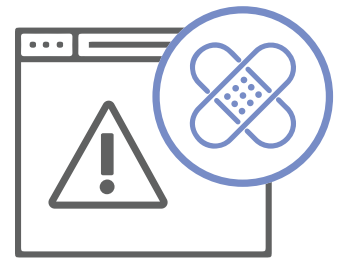
71%

Phishing



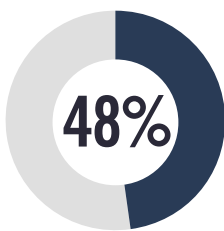
61%

Malware

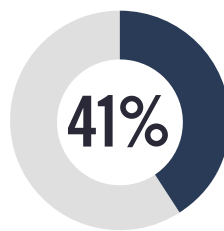


56%

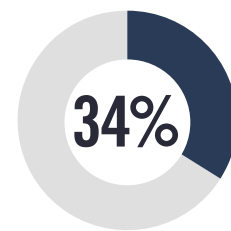
Unpatched systems/
vulnerability exploits



Unauthorized user/
privileged access



Malicious
websites



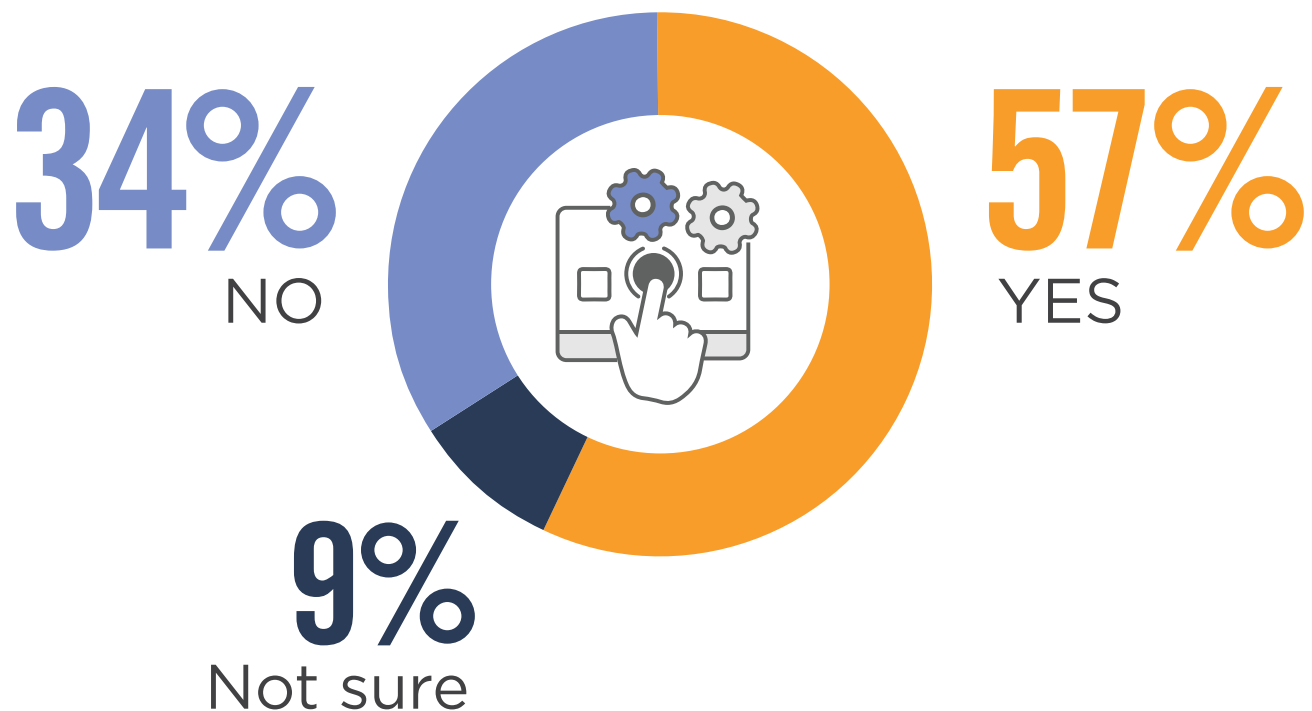
Identity
theft

Insider attacks 20% | Account takeover 20% | Other 1%

ACCESS THROUGH PERSONAL DEVICES

With 57% of workers allowed to access company resources from personal and unmanaged devices, it is no surprise that unpatched systems (56%) and unauthorized access (48%) are high on the list of threat concerns.

► Are employees able to access managed applications from personal, unmanaged devices?



SECURITY SUPPORT ISSUES

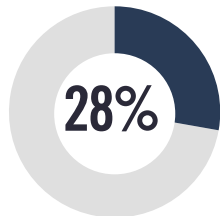
Since people moved to working remotely, two out of three security support cases are about suspected phishing attempts (64%). This is followed by concerns over connecting personal devices and malware, tying at 28%.

▶ Since going remote, which of the following have you received increased security support questions about?

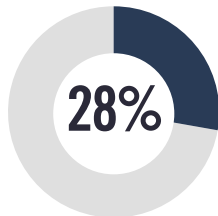


64%

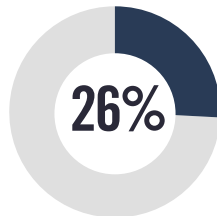
Reported
phishing
attempts



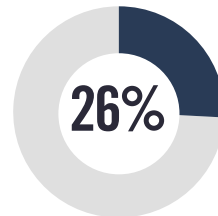
Concerns over
connecting
personal IoT to same
internet network
used for work



Reports of
malware on
devices



Unable to
share/back-up
from certain
device types



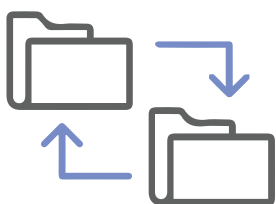
Suspicious
two-factor
authentication
attempts

Reporting breaches of third-party tools 21% | Concerns about terms of service of certain apps 15% | Other 5%

RISKY APPS

When asked what work applications cybersecurity professionals are most concerned about, 66% mention file sharing. This is followed by email and messaging apps (57%) and websites (39%).

► **What work applications used by remote workers are you most concerned about from a security perspective?**



66%

File sharing



57%

Email and messaging



39%

Websites



34%

Social media



32%

Free business web applications



29%

Video conferencing

Subscribed, corporate managed web applications 11% | Other 2%

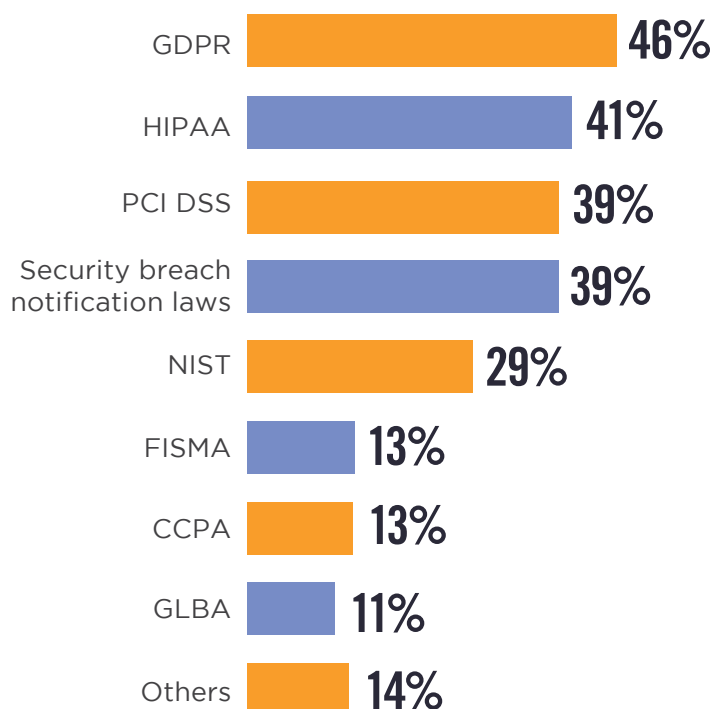
IMPACT ON COMPLIANCE

Sixty-eight percent of organizations agree that remote work could impact compliance posture. The regulations most likely affected include GDPR (46%), HIPAA (41%), PCI DSS (39%), and security breach notification laws (39%).

▶ Could remote work impact compliance mandates that apply to your organization?



▶ If so, which ones?



REMOTE SECURITY

Exposure to malware (42%) and data protection (33%) make up 75% of the concerns regarding people connecting from remote work locations.

► What is the primary risk you're concerned with as your users connect remotely?



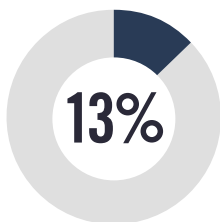
42%

Exposure to malware and phishing risks

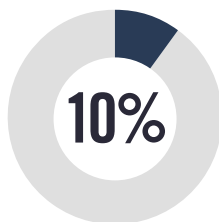


33%

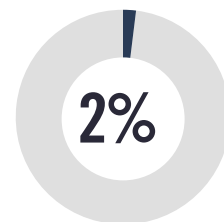
Protection of my data, especially when accessed by unmanaged endpoints



Audit and oversight of employees conducting work from unmanaged resources



Ensure compliance of my regulated users

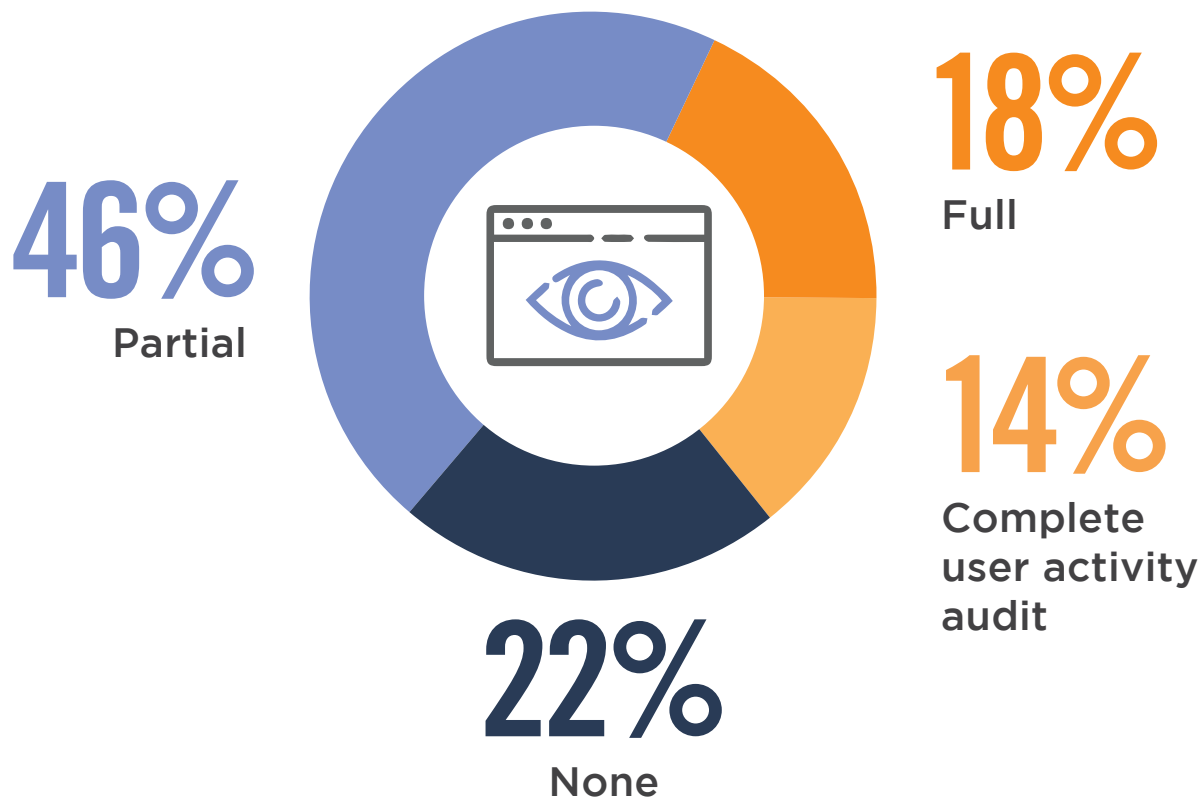


Inability to detect novel attacks across the workforce

VISIBILITY INTO USER ACTIVITY

We asked what visibility organizations have over user activity if VPN is disabled by a remote worker. Only 18% of organizations have full visibility over user activity if VPN is disabled. Sixty-eight percent of organizations have either partial or nonexistent visibility.

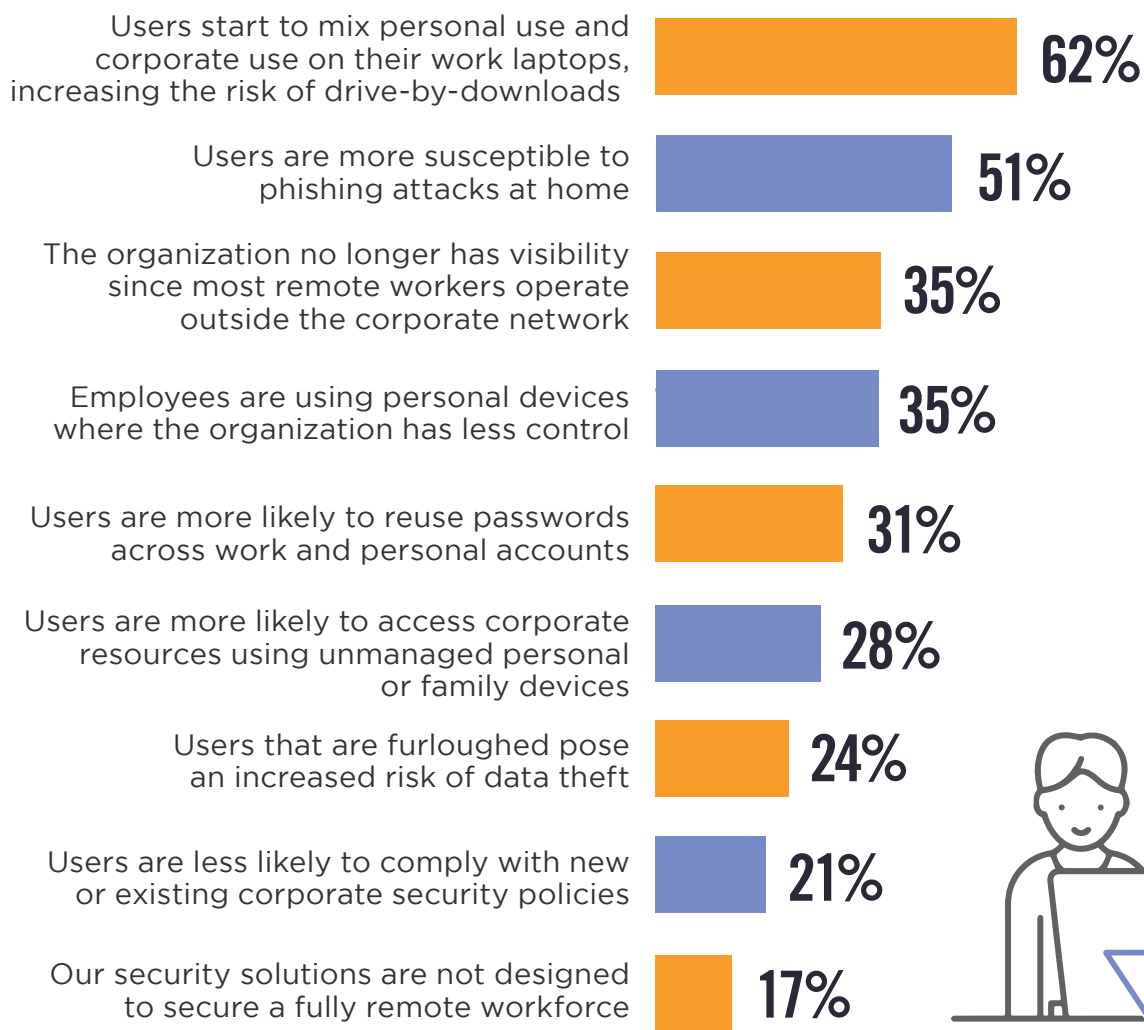
▶ **What visibility do you have over user activity if VPN is disabled by a remote worker?**



WHAT MAKES REMOTE WORK LESS SECURE

One of the main reasons remote work is less secure is because workers are starting to mix personal use and corporate use on their work laptops, increasing the risk of drive-by-downloads (62%). Another reason is that users are also more susceptible to phishing attacks at home (51%). Adding to the risk is that organizations no longer have sufficient visibility as most remote workers operate outside the corporate network (35%).

► What makes remote work less secure?



Security tools cannot adapt to fast-changing environments and user behaviors 2% | We do not have security that covers the entire enterprise 2% | Other 2%

IMPERSONATION & PHISHING ATTACKS CONCERNS

Practically all organizations in our survey (98%) are concerned about sophisticated impersonation and phishing email attacks.

► How concerned is your organization about sophisticated impersonation and/or phishing email attacks?



98% Of organizations are concerned about sophisticated impersonation and/or phishing email attacks.

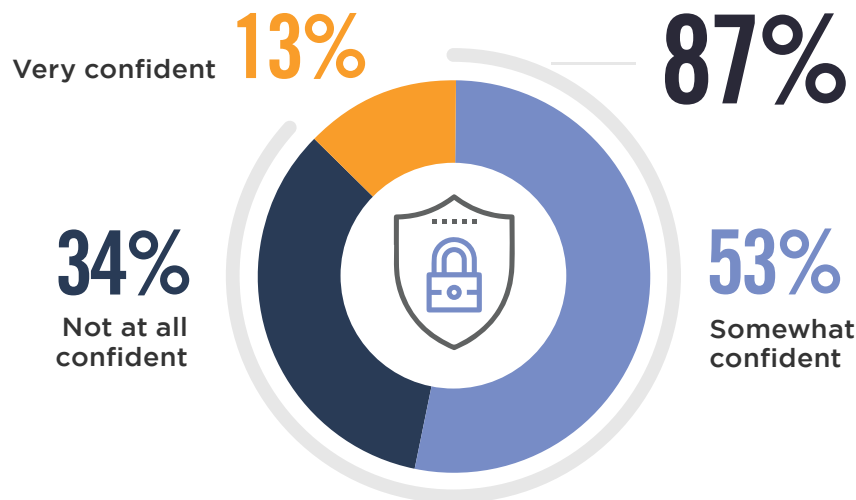


Very concerned Somewhat concerned Not at all concerned

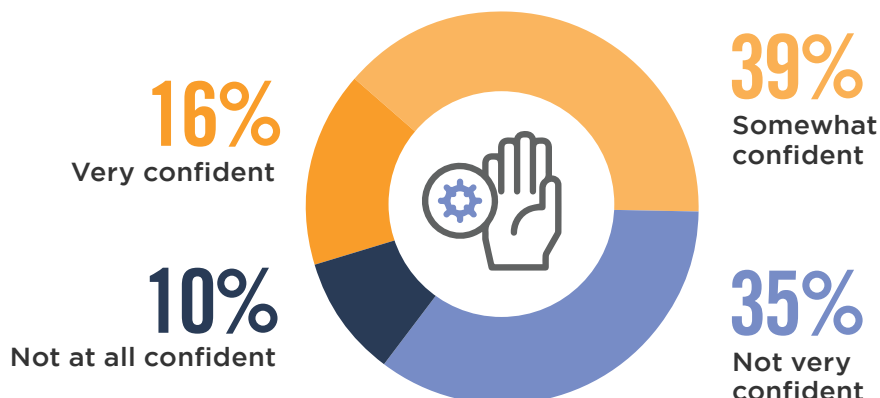
SECURITY TOOLS CONFIDENCE

Confidence is low in the ability of legacy email tools to stop sophisticated security threats. Email attacks are slipping past traditional email security tools with greater frequency, calling for new technologies which can adapt to attackers' tactics and techniques, and autonomously defend their email environment. Eighty-seven percent of security professionals are not confident that legacy email tools can stop a novel email phishing attack from reaching their employees' inboxes. Security professionals are also not confident in their existing solutions with 16% of respondents claiming they are very confident in their existing security tools.

- ▶ **How confident are you that legacy email security tools can stop sophisticated novel email phishing attacks from reaching employee inboxes?**



- ▶ **How confident are you that your existing security tools would be able to identify insider threat or a stealthy attack through the supply chain?**



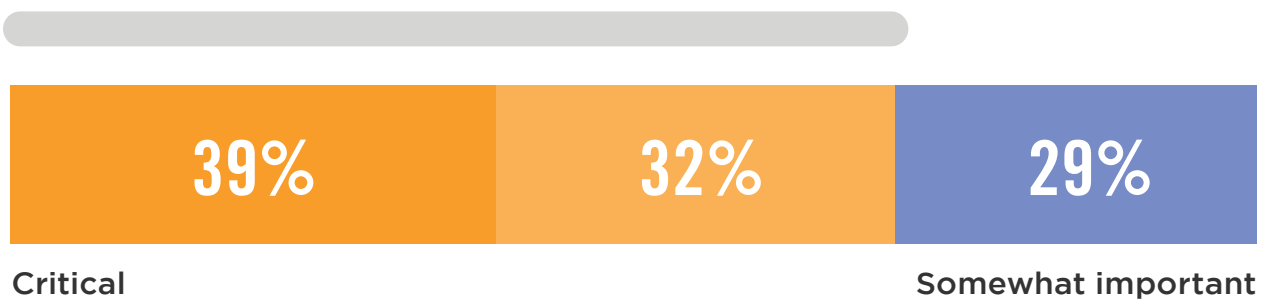
SECURITY TOOLS ADAPTING TO CHANGING ENVIRONMENTS

We asked cybersecurity professionals how important it is that security tools can adapt to the changing digital environment. Seventy-one percent confirm this capability is very to critically important. Rather than rely on fixed rules and signatures, modern security tools should evolve with the dynamic workforce and constantly adapt and respond to stay ahead of cyber-threats. We asked cybersecurity professionals how important it is that security tools can adapt to the changing digital environment. Seventy-one percent confirm this capability is very to critically important.

- ▶ **How important is it that security tools can adapt to changing digital environments, without requiring humans to configure them?**



71% Agree that it's critical to very important for security tools to adapt to the changing digital environment.

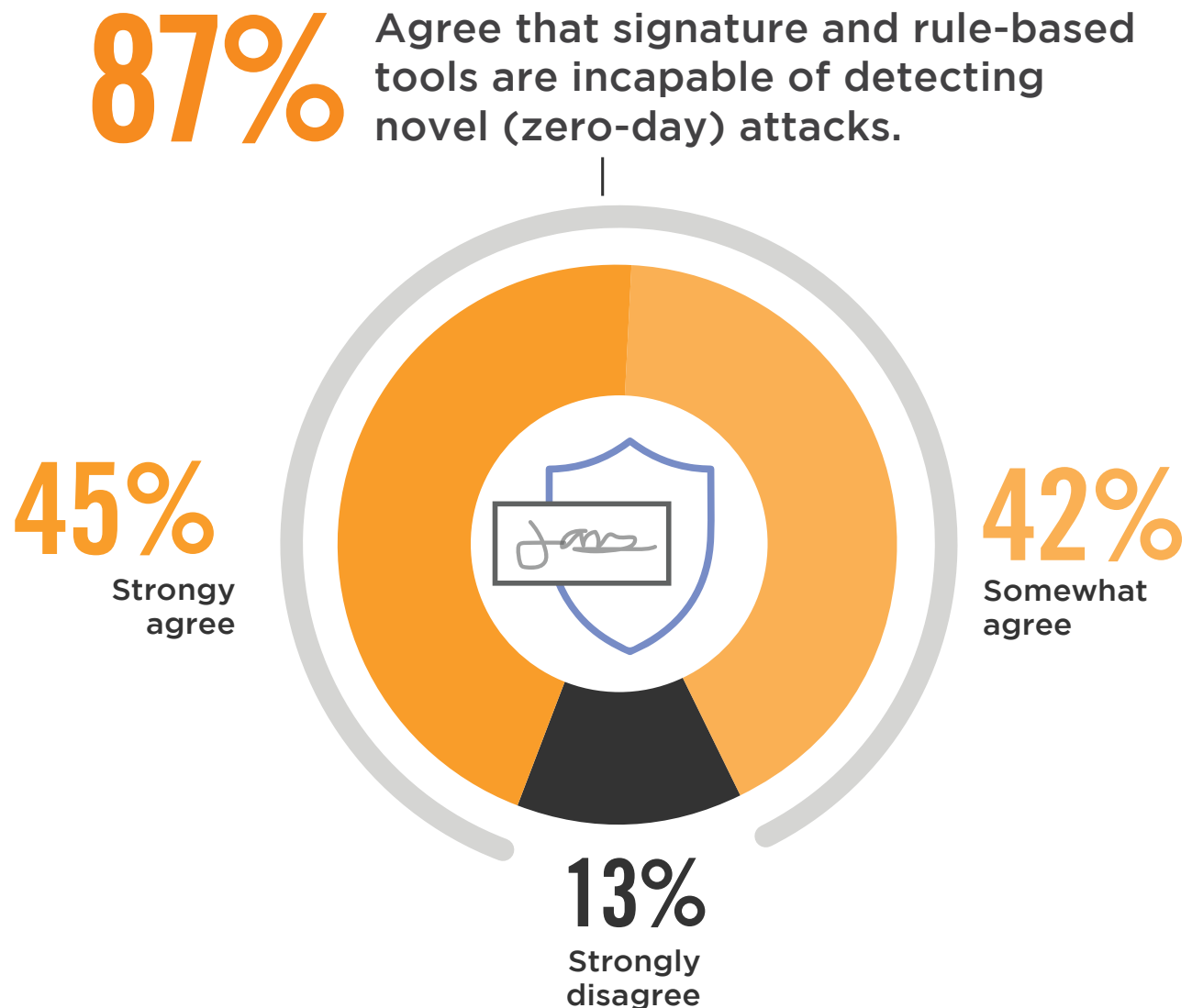


■ Critical ■ Very important ■ Somewhat important

SIGNATURE AND RULE-BASED TOOLS

Eighty-seven percent of security professionals agree that signature and rule-based tools are incapable of detecting novel (zero-day) attacks. As cyber-criminals begin to recognize this limitation of traditional tools and update their attack infrastructure more frequently, the need for a self-learning approach to security will become more critical.

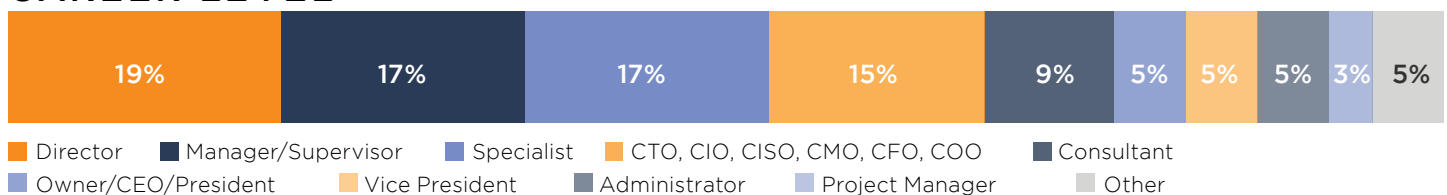
► To what extent do you agree that signature and rule-based tools are incapable of detecting novel (zero-day) attacks?



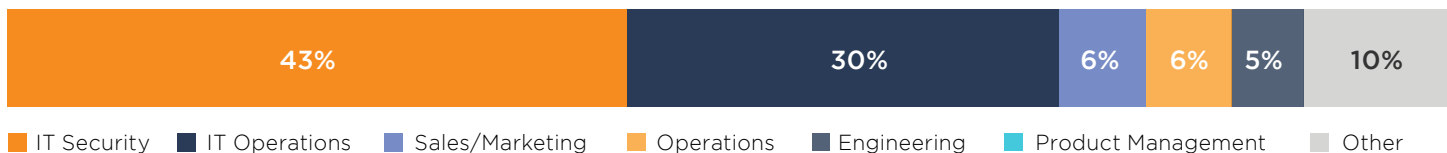
METHODOLOGY & DEMOGRAPHICS

This report is based on the results of a comprehensive online survey of 237 IT and cybersecurity professionals in the US, conducted in March 2021, to identify the latest enterprise adoption trends, challenges, gaps, and solution preferences for remote work security. The respondents range from technical executives to IT security practitioners, representing a balanced cross-section of organizations of varying sizes across multiple industries.

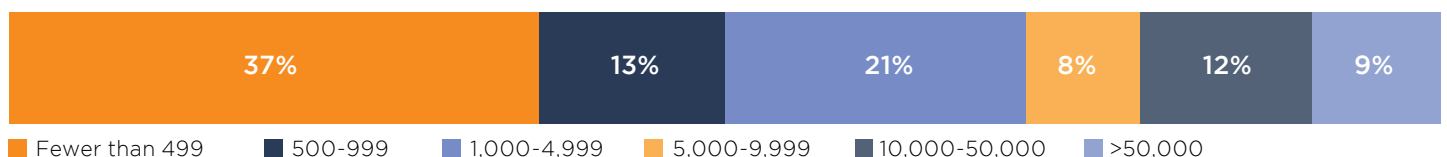
CAREER LEVEL



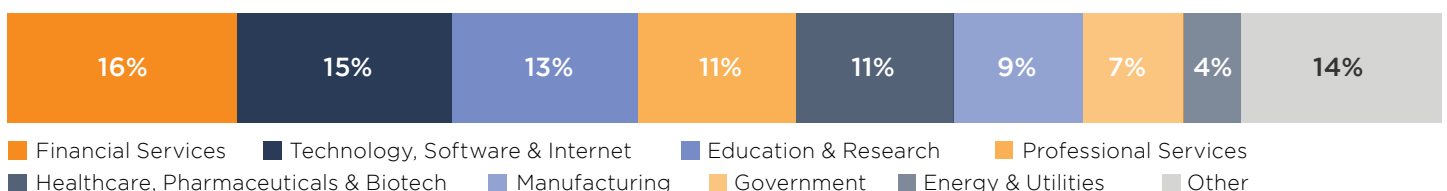
DEPARTMENT



COMPANY SIZE



INDUSTRY





Darktrace is a leading cybersecurity AI company and the creator of Autonomous Response technology. It provides enterprise-wide cyber defense to over 4,700 organizations, protecting the cloud, email, SaaS, traditional networks, IoT devices, endpoints, and industrial systems.

A self-learning technology, Darktrace AI autonomously detects, investigates, and responds to advanced cyber-threats, including insider threat, remote working risks, ransomware, data loss, and supply chain vulnerabilities.

The company has 1,500 employees globally, with headquarters in Cambridge, UK. Every second, Darktrace AI fights back against a cyber-threat, before it can cause damage.

For more information please visit
www.darktrace.com